APPENDICES TO:

"Resilience Before Relapse: A Dynamical Systems Approach to Mobile Depression Monitoring"

Appendix A. Important Notations

Notions	Descriptions
$x_{i,t}$	The monitoring value of behavior *i* at time *t*.
u_T	The dense latent representation used for Riemannian metric calculation.
G_t	The Riemannian metric tensor at time *t*.
$c_{i,t}$	The manifold-aware relative change representation of behavior *i* at time *t*.
$\psi_{ij,t}^{cond}$	The conditional context for the ordered symptom pair (*i, j*), preserving source-specific information from symptom *i*.
$\psi_{ij,t}^{marg}$	The marginal context for the ordered symptom pair (*i, j*), replacing source-specific information with a baseline reference.
$R_{ij}^{(t+1)}$	The unnormalized influence score from symptom *i* to symptom *j*.
$r_{ij,t}$	The normalized directed symptom interplay from symptom *i* to symptom *j*, used to represent system resilience.
$c_{ij,t}$	The joint representation of the symptom pair (*i, j*).
$x_{i,t}^{u}$	The updated symptom representation.
s_T	The overall depression system state.
y_t	The depression state for each individual at time *t*.

Appendix B. Explanation of Riemannian Manifold

In Euclidean space, changes are measured using the same fixed metric. Thus, behavioral change would be measured by its squared Euclidean norm:

$$\|\Delta x_{i,t}\| = \sqrt{\Delta {x_{i,t}}^T \Delta x_{i,t}} \tag{B1}$$

The role of G_t is to adjust the local ruler that changes across the manifold according to the current position on the manifold. In our model, the metric tensor at the current behavioral state $x_{i,t}$ is defined as:

$$G_t = J_\phi(x_{i,t})^T J_\phi(x_{i,t}) \tag{B2}$$

Substituting G_t into the computed manifold-based relative change gives:

$${\Delta x_{i,t}}^T G_t \Delta x_{i,t} = {\Delta x_{i,t}}^T J_\phi(x_{i,t})^T J_\phi(x_{i,t}) \Delta x_{i,t} \tag{B3}$$

A small mapped change on the manifold satisfies:

$$\Delta u_T \approx \frac{\partial u_T}{\partial x_{i,t}} \Delta x_{i,t} = J_\phi(x_{i,t}) \Delta x_{i,t} \tag{B4}$$

It follows that

$$\|\Delta u_T\| = \sqrt{{\Delta u_T}^T \Delta u_T} = c_{i,t} = \|\Delta x_{i,t}\|_{G_t} \tag{B5}$$

Therefore, $c_{i,t}$ represents the behavioral change measured under the local geometry of the learned manifold.

Appendix C. The Descriptions of Behavioral Data

Features	Descriptions
Physical activity	Duration in vehicle / on bike / on foot / still
Audio	Number of audio plays, duration of audio plays
Location	Distance travelled, time spent at home, workout places, study places, social places, dorms, others' dorm, greek houses, max distance from campus, number of locations visited
Phone unlock	Number of phone locks & unlocks, duration of phone unlock
Call	Number of income & outgoing & missing calls, duration of income & outgoing calls
SMS	Number of income & outgoing SMS